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Introduction

"Whatever you would make habitual, practice it; and if you would not make a thing habitual, do not practice it, but accustom yourself to something else." – *Epictetus*

This book is a companion to a course in business statistics. It teaches the core ideas of statistics and how to apply them in Excel. Each chapter introduces a few concepts, works through examples, and ends with practice problems. The material takes effort, but steady work will get you through it.

I owe thanks to my business statistics students, especially Haipei Liu, Addison Alvine, and Sara Blair, for their feedback and careful reviews. Any remaining errors are mine.

Learning from Data

Statistics is about learning from the world. We want to know how customers behave, whether a new process works, or where a market is headed, but we can never observe everything. Instead, we collect **data**, a small and imperfect view of what we care about, and use it to draw conclusions about the bigger picture.

That last step comes with a catch. A sample never perfectly mirrors the population it came from, so two analysts with different samples will get different answers. **Probability** lets us measure how far our data might mislead us. With it, we can test a **hypothesis**, an idea we hold about the world, and decide whether the evidence supports it or whether we are just seeing noise.

You already reason this way in everyday life. You form impressions from limited experience, update them when new evidence arrives, and try not to be fooled by coincidence. Statistics teaches you to do the same thing carefully.

Why Excel?

Excel is the most widely used data tool in business. Finance, accounting, marketing, and operations teams use it every day to organize data, find patterns, and report results. It is available almost everywhere, and the skills you learn here will be useful in your first internship.

Excel can handle every technique in this book, from descriptive statistics and pivot tables to probability, regression, and hypothesis testing. Each chapter shows you how, with step-by-step instructions, screenshots, and downloadable datasets. You do not need any programming experience.

Excel in the Era of AI?

AI tools can now run an analysis in seconds, but you still need to understand the data and the logic to know whether the results make sense. Working in Excel builds that understanding. You learn to organize information, check your assumptions, and follow a calculation from start to finish. These habits make you a better user of AI, because you will be able to question its output instead of simply accepting it.

Roadmap

The book follows the same path as learning from data. **Descriptive Statistics** shows you how to summarize, visualize, and explore a data set. **Regression** introduces tools for measuring relationships between variables and making predictions. **Probability** gives you a way to model the uncertainty that comes from working with samples. **Inference** brings these pieces together so you can reason from a sample to the population and test hypotheses. The final chapter, **Regression and Inference**, uses those tools to judge how reliable a regression model is.

If you are new to spreadsheets, or just want a quick reference, start with the **Excel Basics** chapter at the end of the book. It covers how Excel organizes data (cells, ranges, worksheets, and the Data Analysis ToolPak) and lists every function used in the book.

How to Use This Book

Every chapter has the same structure. It starts with key concepts, formulas, and worked examples. Next, it shows how to carry out each technique in Excel, followed by an **Excel Function Summary**. Many chapters include a dataset you can download to follow along, and the online version adds **Try it** boxes: interactive graphs you can adjust to see a concept in action. Each chapter ends with **Exercises**. Try each problem on your own before clicking to reveal the answer.

How do I get Excel?

William & Mary students can download Office 365, which includes Excel, at https://software.wm.edu/office_365_d1/.

Part I

Describe

1 Descriptive Stats I

Understanding the nature and classification of data is crucial for effective analysis and decision-making. Data are the building blocks of insights, providing a foundation for businesses, researchers, and policymakers to make informed choices. How we collect and categorize data significantly impacts how it is analyzed and interpreted. Below I highlight key types of data and their unique characteristics to help you get started with data analysis. In the arc of this book, this is where learning from data begins: before we can summarize what we have observed or infer what it means for the world, we first have to understand the raw material itself — what kind of data we have collected and how it is organized.

1.1 Data and Types of Data

Let's start by defining data and the types of data structures that you might encounter when conducting analysis. **Data** are facts and figures collected, analyzed and summarized for presentation and interpretation. Data can be classified as:

- **Cross Sectional Data** refers to data collected at the same (or approximately the same) point in time. *Ex: NFL standings in 1980 or Country GDP in 2015.*
- **Time Series Data** refers to data collected over several time periods. *Ex: U.S. inflation rate from 2000-2010 or Tesla deliveries from 2016-2022.*
- **Structured Data** resides in a predefined row-column format (tidy). *Ex: spreadsheet data.*
- **Unstructured Data** does not conform to a pre-defined row-column format. *Ex: Text, video, and other multimedia.*

EXAMPLE

Consider a retail store analyzing its sales performance. If the store collects data on the total revenue generated *by each location* on Black Friday, it is cross-sectional data. On the other hand, if the store tracks *weekly sales for the past year* to observe trends, it is time series data. Structured data, like sales figures stored in *spreadsheets*, allows for easy comparison and analysis. Meanwhile, customer feedback gathered from *social media posts and video reviews* represents unstructured data, requiring advanced tools to extract meaningful insights.

1.2 Data Sets

A **data set** contains all data collected for a particular study. Understanding the structure the data set aids in communicating findings to stakeholders clearly and effectively. Data sets are composed of:

- **Elements** are the entities on which data are collected. *Ex: Football teams, countries, and individuals.*
- **Variables** are a set of characteristics collected for each element. *Ex: Goals scored, GDP, weight.*
- **Observations** are the set of measurements obtained for a particular element. *Ex: Salah, 20 (goals), 15 (assists). US, 2.3 (inflation), 4.5% (federal interest rate).*

Elements	Variable 1	Variable 2
Element 1	#	#
Element 2	#	#
Element 3	#	#
...	...	...

EXAMPLE

Consider the dataset on electric vehicles (EVs) displayed below:

Make	Model	Range_km	TopSpeed_kmh	Price_pounds	Charge_kmh
Tesla	Model 3	415	201	39,990	690
Tesla	Model S	652	250	79,990	750
Nissan	Leaf	270	144	26,995	100
BMW	i3	260	150	37,955	100
Audi	e-tron	400	200	60,000	470
Ford	Mustang	483	180	42,995	750
...	...	...	...	...	...

In this data set, each row represents an electric vehicle model, making the elements the specific EV models rather than the manufacturers. The variables collected for each model include:

- Make: The manufacturer of the EV.
- Model: The specific name of the EV model.

- Range_km: Driving range in kilometers on a full charge.
- TopSpeed_kmh: Maximum speed in km/h.
- Price_pounds: Price in pounds (£).
- Charge_kmh: Charging speed in kilometers per hour.

An example observation is "Tesla Model 3," with the following data: Make: Tesla, Model: Model 3, Range_km: 415, TopSpeed_kmh: 201, Price_pounds: 39,990, Charge_kmh: 690.

1.3 Scales of Measurement

Understanding scales of measurement is crucial for analyzing and interpreting data effectively in business. By distinguishing between categorical (e.g., marital status, satisfaction ratings) and numerical data (e.g., profits, prices), you'll know what methods to use for analysis. Knowing whether data is nominal, ordinal, interval, or ratio ensures your analysis and conclusions are accurate and relevant.

The **scales of measurements** determine the amount and type of information contained in each variable. In general, variables can be classified as **categorical** or **numerical**.

- **Categorical** (qualitative) data includes labels or names to identify an attribute of each element. Categorical data can be **nominal** or **ordinal**.
  - With **nominal** data, the order of the categories is arbitrary. Numbers (if used) serve only as identifiers. Common statistics used are based on counting frequencies and the mode (most common category) *Ex: Marital Status, Race/Ethnicity, or NFL division.*
  - With **ordinal** data, the order or rank of the categories is meaningful, but intervals between ranks are not necessarily equal or quantifiable. Common statistics are based on median, percentiles, or rank-order correlations. *Ex: Rating, Difficulty Level, or Spice Level.*
- **Numerical** (quantitative) include numerical values that indicate how many (discrete) or how much (continuous). The data can be either **interval** or **ratio**.
  - With **interval** data, the distance between values is expressed in terms of a fixed unit of measure. The zero value is arbitrary and does not represent the absence of the characteristic. Ratios are not meaningful. Common statistics are the arithmetic mean, standard deviation, or the Pearson correlation, *Ex: Temperature or Dates.*
  - With **ratio** data, the ratio between values is meaningful. The zero value is not arbitrary and represents the absence of the characteristic. All statistics are permissable including geometric mean, and the coefficient of variation. *Ex: Prices, Profits, Wins.*

EXAMPLE

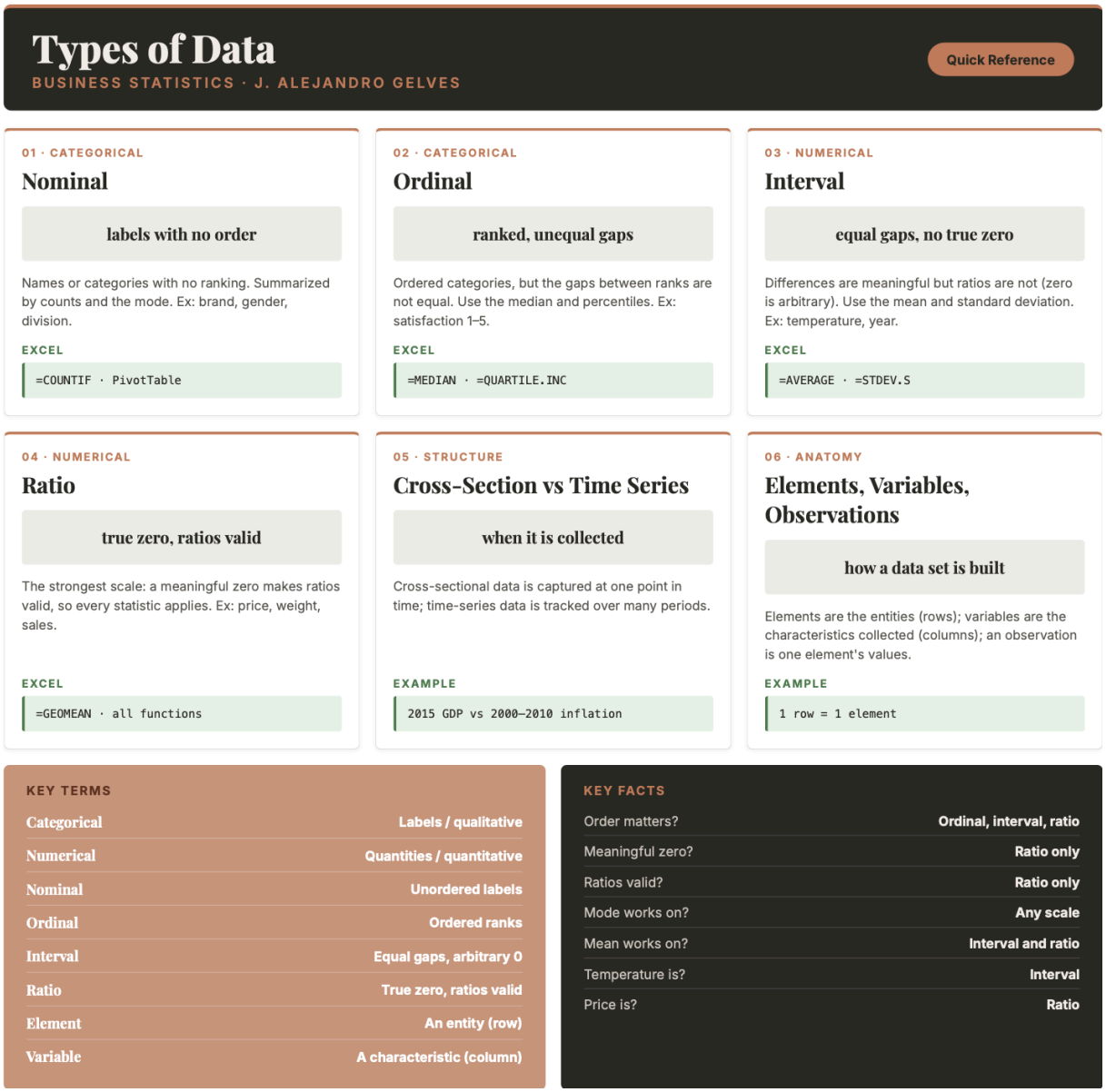
Let's keep using the EV example. Consider the new data set below:

Car	Brand	Range	Rating	Year
Model 3	Tesla	415	4	2021
Model S	Tesla	652	5	2021
Leaf	Nissan	270	3	2021
i3	BMW	260	3	2021
e-tron	Audi	400	4	2021
Mustang	Ford	483	4	2021
...	...	...	...	...

The variables can be classified as follows:

- Car (Categorical - Nominal), consists of names of cars, which are labels used to identify each row. The order of these names does not matter, making it nominal data.
- Brand (Categorical - Nominal) represents the manufacturer of the car (e.g., Ford, Audi). These are labels with no inherent order, making it nominal data.
- Range (Numerical - Ratio), refers to the car's driving range in miles. It is numerical and ratio because it has a meaningful zero (a car with zero range cannot move), and ratios are meaningful (e.g., a car with 250 miles range has double the range of one with 125 miles).
- Rating (Categorical - Ordinal) represents a rank or score (e.g., 4, 3, 2). The order matters, as higher ratings indicate better performance. However, the intervals between ratings are not consistent, so it is ordinal data.
- Year (Numerical - Interval) represents a point in time. While numerical, it is interval data because the zero point is arbitrary (e.g., year 0 does not indicate the "absence" of time), and ratios are not meaningful (e.g., 2020 is not "twice as late" as 1010).

1.4 Chapter Summary Cheat Sheet



1.5 Exercises

The following exercises will help you test your knowledge on the Scales of Measurement. In these exercises you will:

- Identify numerical and categorical data.
- Classify data according to their scale of measurement.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you formulate your own answer and double checked your work for any mistakes.

Exercise 1

A bookstore has compiled data set on their current inventory. A portion of the data is shown below:

Title	Price	Year Published	Rating
Frankenstein	5.49	1818	4.2
Dracula	7.60	1897	4.0
...	...	...	...
Sleepy Hollow	6.95	1820	3.8

1. Which of the above variables are categorical and which are numerical?

Answer on page ?? →

2. What is the measurement scale of each of the above variable?

Answer on page ?? →

Exercise 2

A car company tracks the number of deliveries every quarter. A portion of the data is shown below:

Year	Quarter	Deliveries
2016	1	14800
2016	2	14400
...	...	...
2022	3	343840

1. What is the measurement scale of the Year variable? What are the strengths and weaknesses of this type of measurement scale?

Answer on page ?? →

2. What is the measurement scale for the Quarter variable? What is the weakness of this type of measurement scale?

Answer on page ?? →

3. What is the measurement scale for the Deliveries variable? What are the strengths of this type of measurement scale?

Answer on page ?? →

Exercise 3

A coffee shop chain collects customer feedback data for analysis. For each visit/transaction recorded in their database, the following variables are stored:

- Customer ID: A unique identifier assigned to each registered customer (e.g., C001234, C005678).
- Satisfaction Level: Customers rate their overall experience as 1 (Very Dissatisfied), 2 (Dissatisfied), 3 (Neutral), 4 (Satisfied), or 5 (Very Satisfied).
- Visit Duration: The time spent in the store in minutes (e.g., 15, 42, 8).
- Beverage Type: The main drink ordered (e.g., Latte, Cappuccino, Americano, Tea, Other).

Answer on page ?? →